

Resilience Under Pressure

A Data-Driven Framework for Global Disaster Risk, Response, and Recovery Analytics

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Abstract—This is a one-paragraph abstract summarizing the problem, methods used, and key findings.

I. PROBLEM STATEMENT AND OBJECTIVES

A. Problem Statement

In recent decades, the global frequency of natural and man-made disasters ranging from catastrophic floods and wildfires to pandemics has escalated significantly. These events expose critical vulnerabilities in human systems, highlighting governance failures, economic fragility, and infrastructure weaknesses. Currently, disaster management is predominantly reactive, focusing on immediate relief rather than long-term systemic resilience.

A critical gap exists in the ability of global organizations, such as the Global Disaster and Humanitarian Response Agency (GDHRA), to quantitatively measure “resilience.” While vast amounts of data exist regarding disaster events (e.g., EM-DAT) and socio-economic indicators (e.g., World Bank, UNDP), these datasets remain unstructured and disconnected. Decision-makers lack a unified, data-driven framework to assess why certain nations absorb shocks effectively while others face prolonged collapse. Without a computational approach to define resilience, it is difficult to identify the specific drivers—be they economic, institutional, or environmental—that facilitate rapid recovery.

B. Objectives

The primary goal of this project is to design a prototype analytical platform for the GDHRA that transforms unstructured public data into actionable intelligence. By integrating historical disaster data with socio-economic indicators, this project aims to quantify and visualize resilience at national and regional scales.

The specific objectives of this study are:

- **Data Fusion and Integration:** To aggregate and harmonize heterogeneous datasets from sources such as EM-DAT, the World Bank, and the UNDP, aligning them by country and year to create a unified dataset for analysis.
- **Quantitative Modeling of Resilience:** To formulate and implement computational metrics for resilience, specifically calculating the Disaster Impact Index (DII), Resilience Recovery Score (RRS), and Composite Resilience Index (CRI). These models aim to mathemati-

cally represent the relationship between disaster exposure, vulnerability, and adaptive capacity.

- **Feature Engineering:** To derive critical variables including binary disaster frequency, per capita impact metrics (deaths, affected population, and economic damage), the adaptive capacity index, and a normalized disaster severity index, to facilitate the calculation of resilience models.
- **Interactive Visualization:** To develop an exploratory Tableau dashboard that allows decision-makers to analyze trends across time, geography, and disaster types, facilitating the identification of high-risk versus high-resilience nations.
- **Analytical Insight:** To determine the correlation between income stability, economic wealth (GDP), and disaster recovery metrics, thereby providing evidence-based recommendations for improving global resilience.

II. DATA SOURCES AND PREPROCESSING STEPS

To construct a holistic view of global resilience, we aggregated data from six distinct sources covering the period 1900–2024. The final processed dataset comprises 79,430 country-year records across 254 unique geographical entities.

A. Data Acquisition

The data collection strategy focused on four domains: Disaster History, Climate Risks, Economic Stability, and Human Development. Table I summarizes the primary datasets utilized.

TABLE I: Summary of Primary Data Sources

Domain	Source	Period	Format
Disasters	CRED (EM-DAT)	1900–2024	.xlsx, .csv
Climate	Our World in Data	1900–2023	.csv
Economy	World Bank (WDI)	1900–2024	.csv
Population	World Bank (WDI)	1900–2024	.csv
Development	UNDP (HDI)	1900–2023	.csv

B. Data Integration and Fusion

The raw data existed in heterogeneous formats (CSV and Excel) with inconsistent country naming conventions. To unify

these sources, we implemented a data fusion pipeline using pandas. The primary challenge was aligning country identifiers. We utilized standard ISO-3166 Alpha-3 codes as the primary key. A mapping process corrected 4,330 records with missing or non-standard ISO codes (e.g., historical entity names), ensuring a successful merge rate.

C. Cleaning and Imputation Strategy

Given the historical nature of the data (spanning over a century), missing values were a significant challenge. We applied domain-specific cleaning logic to ensure analytical validity:

1) *Handling Disaster Data:* The EM-DAT dataset only records events that occurred. Missing values in disaster columns (e.g., Total Deaths, Total Damage) were assumed to represent non-occurrence. Therefore, these null values were imputed with 0.

2) *Interpolation of Socio-Economic Metrics:* Unlike disaster data, missing economic or climate data implies gaps in reporting rather than a lack of activity.

- **Linear Interpolation:** We applied linear interpolation strictly *per country* to fill gaps in time series data (e.g., GDP, Population, HDI).
- **Outlier Removal:** Extreme anomalies in GDP growth were detected; 54 records exceeding realistic economic volatility (e.g., caused by hyperinflation data errors) were flagged and capped to a range of $[-10, 10]$ for the recovery rate calculation.

D. Final Dataset Specifications

The preprocessing pipeline yielded three distinct dataset versions:

- 1) **Raw Merged:** The initial joining of all sources.
- 2) **Cleaned:** The production-ready dataset with 3.16% or fewer missing values in key columns.
- 3) **Feature-Enhanced:** The final analytical set containing 27 columns, including engineered features such as `deaths_per_100k` and `adaptive_capacity_index`.

III. METHODS AND IMPLEMENTATION DETAILS

A. Computational Environment

The analytical framework was developed using Python (v3.6+) in a local Windows execution environment. The core tech stack relies on pandas for extensive data manipulation and merging of CSV and Excel datasets (utilizing `openpyxl` as a dependency), while numpy handles numerical operations.

The project follows a modular directory structure to separate data fusion from model application:

- **Data Fusion:** The entry script, `merge_datasets.py`, orchestrates the integration of disparate data sources (Climate, Disasters, GDP, HDI) into a single country-year panel dataset spanning the years 1900–2025.
- **Feature Engineering:** The `FeatureEngr/feature_engineering.py` module processes the merged data

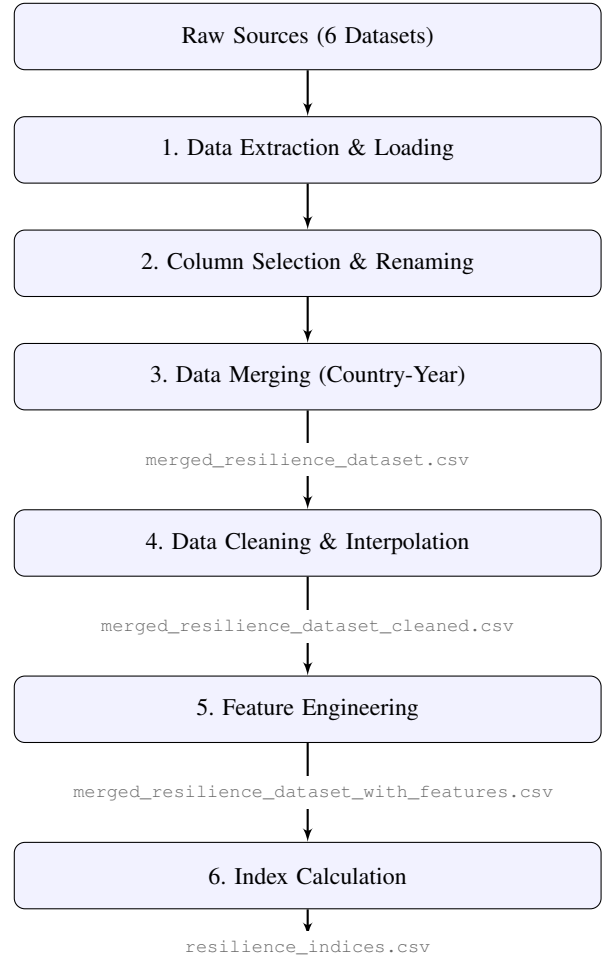


Fig. 1: The data processing pipeline illustrating the transformation from raw sources to calculated resilience indices.

to compute complex resilience metrics, specifically implementing the logic for the Resilience Recovery Score (RRS) and Composite Resilience Index (CRI).

Configuration parameters were managed via JSON files (e.g., `dataset_columns.json`) to ensure reproducibility and clean separation between code and configuration.

B. Mathematical Formulation of Indices

To quantify resilience, we implemented three core indices based on the GDHRA framework. These models integrate derived features, specifically disaster frequency, severity, and adaptive capacity, to produce actionable metrics for country rankings and temporal trend analysis.

1) *Disaster Impact Index (DII):* The DII measures the severity of disaster impact on a country, accounting for both human and economic costs relative to economic capacity. It is calculated as:

$$DII = \left(\frac{F + A_{pop}}{GDP_{pc}} \right) \times S_w \quad (1)$$

Where:

- F : Fatalities per million people.
- A_{pop} : Affected population percentage.
- GDP_{pc} : GDP per capita (scaling factor for economic capacity).
- S_w : Severity weight (based on disaster type and intensity).

Interpretation: A higher DII indicates a greater disaster impact. By normalizing against GDP per capita, this metric highlights nations where disasters cause disproportionate strain relative to their economic size.

2) *Resilience Recovery Score (RRS)*: The RRS estimates a country's capacity to recover, combining economic rebound data with social development indicators. It is defined as the mean of recovery rate, Human Development Index (HDI), and adaptive capacity:

$$RRS = \frac{R_{rate} + HDI + A_{cap}}{3} \quad (2)$$

Where:

- R_{rate} : Calculated recovery rate (normalized).
- HDI : Human Development Index.
- A_{cap} : Adaptive Capacity Index.

Interpretation: The score ranges from 0 to 1. A higher RRS implies superior recovery capacity. This metric balances immediate economic bounce-back with long-term structural strengths like health and education.

3) *Composite Resilience Index (CRI)*: The CRI is the comprehensive measure of a country's overall ability to withstand and recover from shocks. It is modeled as the ratio of capacity to risk:

$$CRI = \frac{A_{cap}}{E \times V} \quad (3)$$

Where:

- A_{cap} : Adaptive Capacity (composite of HDI, institutional stability, and infrastructure).
- E : Disaster Exposure (frequency $\times$ intensity).
- V : Vulnerability (impact per capita).

Interpretation: A higher CRI indicates greater overall resilience. The highest scores are achieved by nations with high adaptive capacity combined with managed exposure and low vulnerability.

4) *Derived Dataset Specifications*: The implementation yields a final analytical dataset (`resilience_indices.csv`) containing:

- **Identifiers**: Country Name, ISO Code, Year.
- **Metrics**: DII, RRS, CRI.

This structured output is designed for immediate application in comparative studies, temporal trend analysis, and dashboard visualization.

C. Feature Normalization and Scaling

To ensure comparability across nations with vastly different demographics and economic profiles, specific normalization techniques were applied to input variables.

1) *Normalization of Component Indices*: Intermediate indices were scaled to a [0, 1] range to serve as consistent building blocks for the primary models:

- **Adaptive Capacity Index**: Calculated as the average of the HDI, Income Index, and normalized Life Expectancy (LE). Life expectancy was explicitly normalized using the standard actuarial formula:

$$LE_{norm} = \frac{LE - 20}{85 - 20} \quad (4)$$

- **Disaster Severity Index**: Derived via a weighted average of Min-Max scaled death counts, affected population proportions, and economic damage.

- **Disaster Exposure**: Computed as the product of annualized disaster frequency (5-year rolling mean) and the normalized severity index.

2) *Scaling of Primary Resilience Indices*: The final resilience metrics underwent explicit scaling to facilitate ranking:

- **CRI**: Final values were Min-Max scaled to [0, 1] after calculating the ratio of adaptive capacity to risk.
- **RRS**: Constructed as the arithmetic mean of three normalized components: normalized recovery, HDI, and adaptive capacity.

3) *Handling of Outliers and Non-Normalized Features*: Certain inputs required specific handling rather than standard [0, 1] scaling to preserve analytical nuance:

- `recovery_rate`: To mitigate the impact of extreme economic shocks or booms, this variable was clipped to the range $[-10, 10]$ rather than normalized to unity.
- `vulnerability_index`: Retained as a weighted average of human and economic vulnerability. Unlike the CRI, this index is not strictly bounded at 1.0, allowing it to capture extreme vulnerability scenarios without compression.

IV. KEY RESULTS, VISUALIZATIONS, AND INTERPRETATION

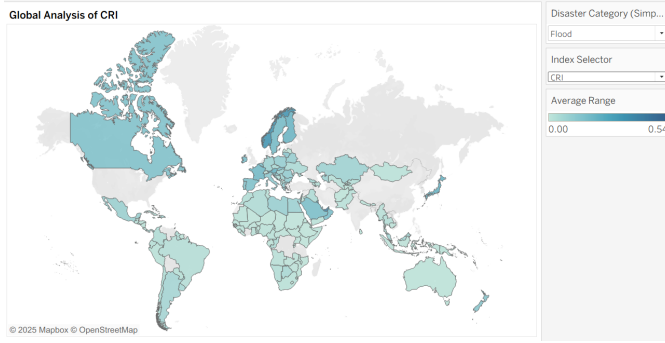
The analytical platform reveals distinct patterns in how nations absorb and recover from shocks. The following results were derived from the interactive Tableau dashboard, structured to address four core analytical questions.

A. *RQ1: Which nations show high exposure but low vulnerability?*

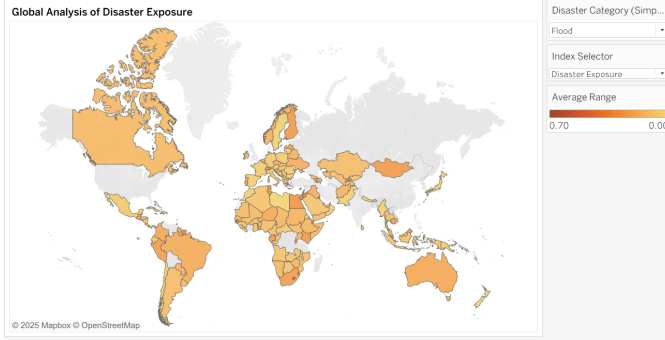
To answer this, we designed a **Toggle-Based Choropleth Map** (see Figure ??).

Design Rationale: The visualization prioritizes lowest cognitive load by strictly separating concepts. Adhering to Tufte's principle of "one map, one meaning," we utilized a parameter-based toggle. This avoids conflicting visual metaphors, ensuring that stakeholders can instantly decode "Risk" versus "Resilience."

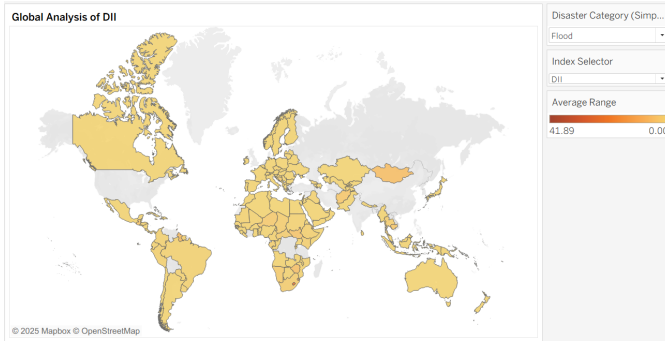
- **Observation:** When toggling between *Disaster Exposure* (Fig. 2b) and *CRI* (Fig. 2a), a clear functional divergence appears. Several high-income nations exhibit extreme



(a) Composite Resilience Index (CRI)



(b) Disaster Exposure



(c) Disaster Impact Index (DII)

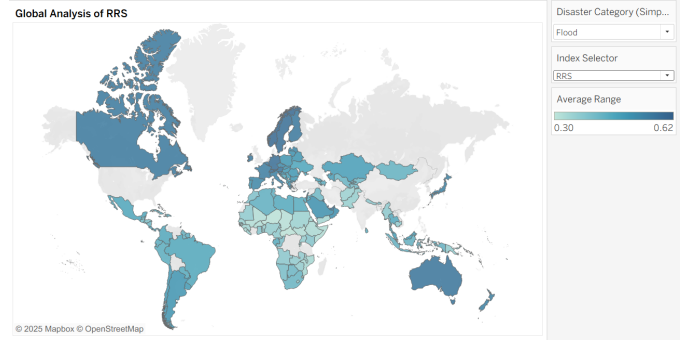
Fig. 2: Geospatial Analysis Part I: (a) Resilience, (b) Exposure, and (c) Impact. Note how high-income nations maintain resilience despite high exposure.

exposure to natural hazards yet maintain high CRI scores. This suggests that advanced infrastructure effectively decouples "Exposure" from "Vulnerability."

- **The Resilience Divide:** A "North-South" divide is evident. Regions in the Global North consistently maintain CRI scores > 0.50 regardless of their risk profile, while the Global South shows high vulnerability even in areas with only moderate exposure.

B. RQ2: How does resilience evolve alongside climate risk?

Figure 4 visualizes the temporal evolution of the Resilience Recovery Score (RRS) across the income groups defined in Section III-D.



(a) Resilience Recovery Score (RRS)

Fig. 3: Geospatial Analysis Part II: (d) Recovery Capacity matches the resilience patterns seen in Part I.

- **Climate Coupling:** The shaded area represents the rising average temperature change from GHGs. Since 1990, climate risk has escalated exponentially.
- **The Adaptation Gap:** High-Income nations (Green line) show a steady upward trend in resilience ($0.48 \rightarrow 0.53$), effectively shielding their economies from climate shocks. In contrast, Under-Developed nations (Red line) have remained stagnant (≈ 0.30) for seven decades.

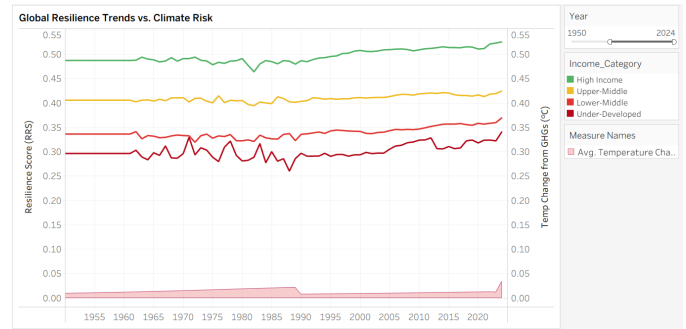


Fig. 4: Global Resilience Trends (RRS) vs. Climate Risk. The widening gap between the Green line (High Income) and Red line (Low Income) illustrates that wealth is the primary buffer against rising temperatures.

C. RQ3: Does National Wealth Guarantee Disaster Resilience?

To test the hypothesis that money equals resilience, we analyzed the correlation between Income Index and RRS (Figure 5).

- **Income as a Predictor:** The scatter plot reveals a strong positive linear relationship. Wealthier nations possess the fiscal space to fund rapid recovery, confirming income as the dominant driver of the RRS.
- **The Exposure Drag:** However, wealth is not a total guarantee. The bubbles are sized by *Disaster Exposure*. Notably, highly exposed nations (large bubbles) frequently sit *below* the regression line. This indicates an "Exposure Drag": extreme disaster frequency suppresses

the recovery potential that a country's wealth would otherwise predict.

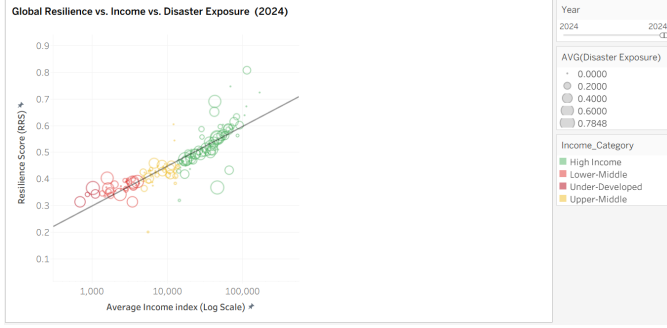


Fig. 5: Scatter plot of Resilience (RRS) vs. Income Index. Bubble size = Disaster Exposure. While wealth correlates with resilience, high exposure (large bubbles) depresses performance below the expected trend.

D. RQ4: Which countries bear the heaviest burden?

The Treemap analysis (Figure 6) visualizes the cumulative Disaster Impact Index (DII), allowing for hyper-localized analysis of burden.

- **Disproportionate Impact:** The visualization confirms that the "heaviest burden" is not carried by the largest global economies, but by smaller developing nations. When filtered for specific disaster types like Floods, we observe that lower-income economies suffer disproportionately high impact scores relative to their GDP.
- **Hazard Specificity:** The view highlights that impact is highly localized; while hazards may be distributed globally, the lasting economic scar is deepest in nations with low adaptive capacity.

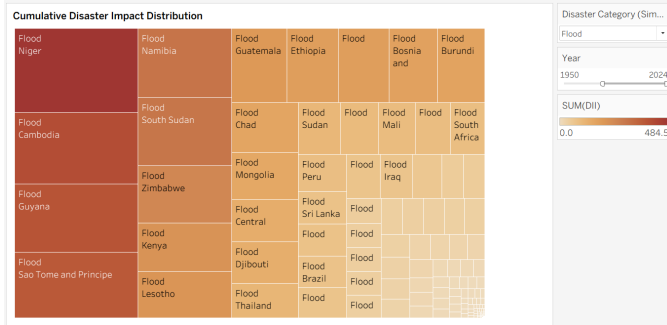


Fig. 6: Cumulative Disaster Impact Distribution (DII). The area represents the magnitude of impact relative to GDP, highlighting the fragility of specific developing economies.

V. LIMITATIONS AND FUTURE WORK

A. Data Limitations

While the framework integrates multi-domain data, several constraints persist:

- **Historical Gaps:** Reliable socio-economic data (GDP, HDI) prior to 1990 is sparse for developing nations.

Consequently, our reliance on linear interpolation for the 1900–1990 period may smooth over short-term economic shocks, potentially underestimating volatility in early decades.

- **Governance Data Exclusion:** although the Worldwide Governance Indicators (WGI) were initially targeted, the dataset only begins in 1996. To maintain a consistent long-term time series (1950–2024), we excluded governance as a direct variable in the composite CRI, relying instead on Income Index as a proxy for institutional capacity.
- **Reporting Bias:** The EM-DAT database relies on self-reporting. Smaller disasters in remote regions may be under-reported compared to major events in developed nations, potentially skewing the Exposure metric toward well-monitored regions.

B. Methodological Constraints

The linear nature of the indices (DII, RRS) assumes equal weighting of components unless specified otherwise. In reality, the interaction between disaster impact and recovery is non-linear—a "tipping point" likely exists where infrastructure collapse makes recovery impossible, which our current linear models do not fully capture. Additionally, the clipping of recovery rates to $[-10, 10]$ prevents extreme outliers from breaking the visualization but may mask true black-swan economic events.

C. Future Directions

To evolve this prototype into a production-grade system, future work should focus on:

- **Predictive Analytics:** Integrating Machine Learning (e.g., LSTM or Random Forests) to forecast future CRI scores based on climate change projections (RCP scenarios).
- **Sub-National Granularity:** Disasters are often localized. Moving from country-level to geospatial grid-level analysis would prevent large nations (like the US or China) from appearing "safe" on average despite having highly vulnerable specific regions.
- **Real-Time Data Pipelines:** Replacing static CSV uploads with API connections to the World Bank and ReliefWeb to provide real-time situational awareness for active disaster events.

APPENDIX

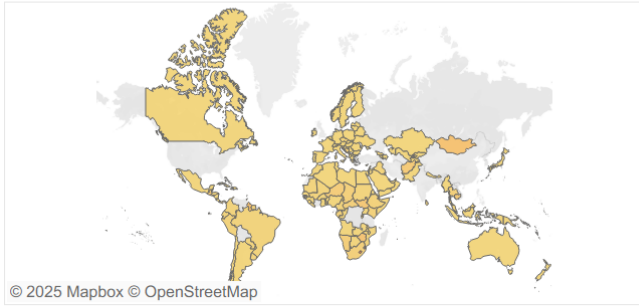
To ensure reproducibility, Table II details the primary and derived variables used in the final dataset (`merged_resilience_dataset_with_features.csv`).

Figure 7 illustrates the complete user interface of the analytical platform. The design integrates the geospatial view (left) with temporal trends and comparative scatter plots (right). A global parameter toggle allows users to switch between viewing Resilience, Risk, and Recovery metrics dynamically.

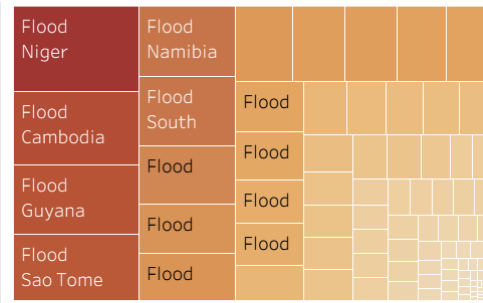
The core resilience indices were calculated using the following logic in `feature_engineering.py`:

Global Climate Risk & Resilience Monitor

Global Analysis of DII



Cumulative Disaster Impact Distribution



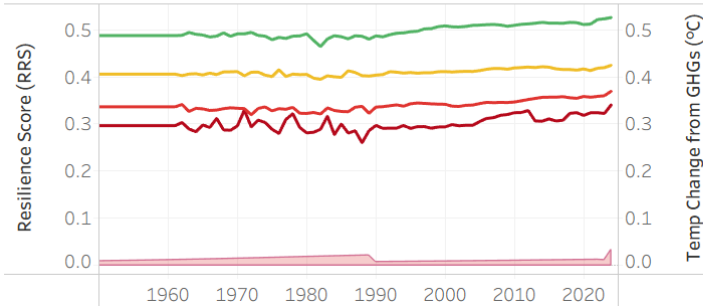
Year
1950 2024

Index Selector
DII

Disaster Categor..
Flood

Income_Category
High Income
Upper-Middle
Lower-Middle
Under-Developed

Global Resilience Trends vs. Climate Risk



Global Resilience vs. Income vs. Disaster Exposure (2024)

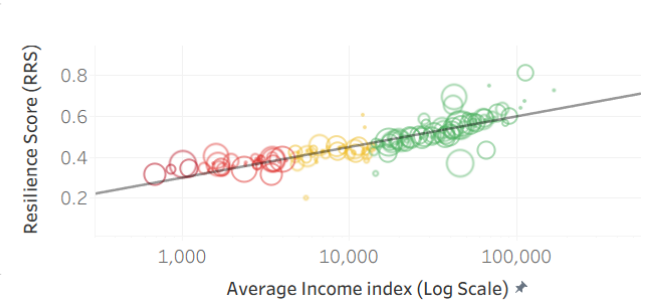


Fig. 7: The Global Resilience Analytics Dashboard. The interactive layout features a control panel for metric selection (CRI, DII, RRS), enabling real-time comparative analysis across geography and time.

TABLE II: Data Dictionary of Key Analytical Variables

Variable	Description
CRI	Composite Resilience Index (0-1). The ratio of adaptive capacity to risk exposure.
RRS	Resilience Recovery Score (0-1). Average of recovery rate, HDI, and adaptive capacity.
DII	Disaster Impact Index. Normalized score of deaths and damage relative to GDP.
disaster_exposure	Product of annualized disaster frequency and normalized severity intensity.
adaptive_capacity	Composite average of HDI, Income Index, and normalized Life Expectancy.
recovery_rate	The delta between post-disaster and pre-disaster GDP growth (clipped to $[-10, 10]$).
Income_Category	Socio-economic binning based on World Bank Atlas method (High, Upper-Middle, Lower-Middle, Low).

```
1. Resilience Recovery Score (RRS)
# Logic: Average of Recovery, HDI, and Capacity
df['RRS'] = (
    df['norm_recovery'] +
    df['HDI'] +
    df['adaptive_capacity_index']
) / 3
```

```
# 2. Composite Resilience Index (CRI)
# Logic: Capacity divided by (Exposure * Vulnerability)
# Small epsilon (1e-6) added to prevent division by zero
risk_factor = (
    df['disaster_exposure'] * df['norm_vulnerability']
```

```
)
df['CRI'] = df['adaptive_capacity_index'] / (risk_factor + 1e-6)

# 3. Normalization (Min-Max Scaling)
scaler = MinMaxScaler()
df[['CRI', 'RRS']] = scaler.fit_transform(
    df[['CRI', 'RRS']]
)
```

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